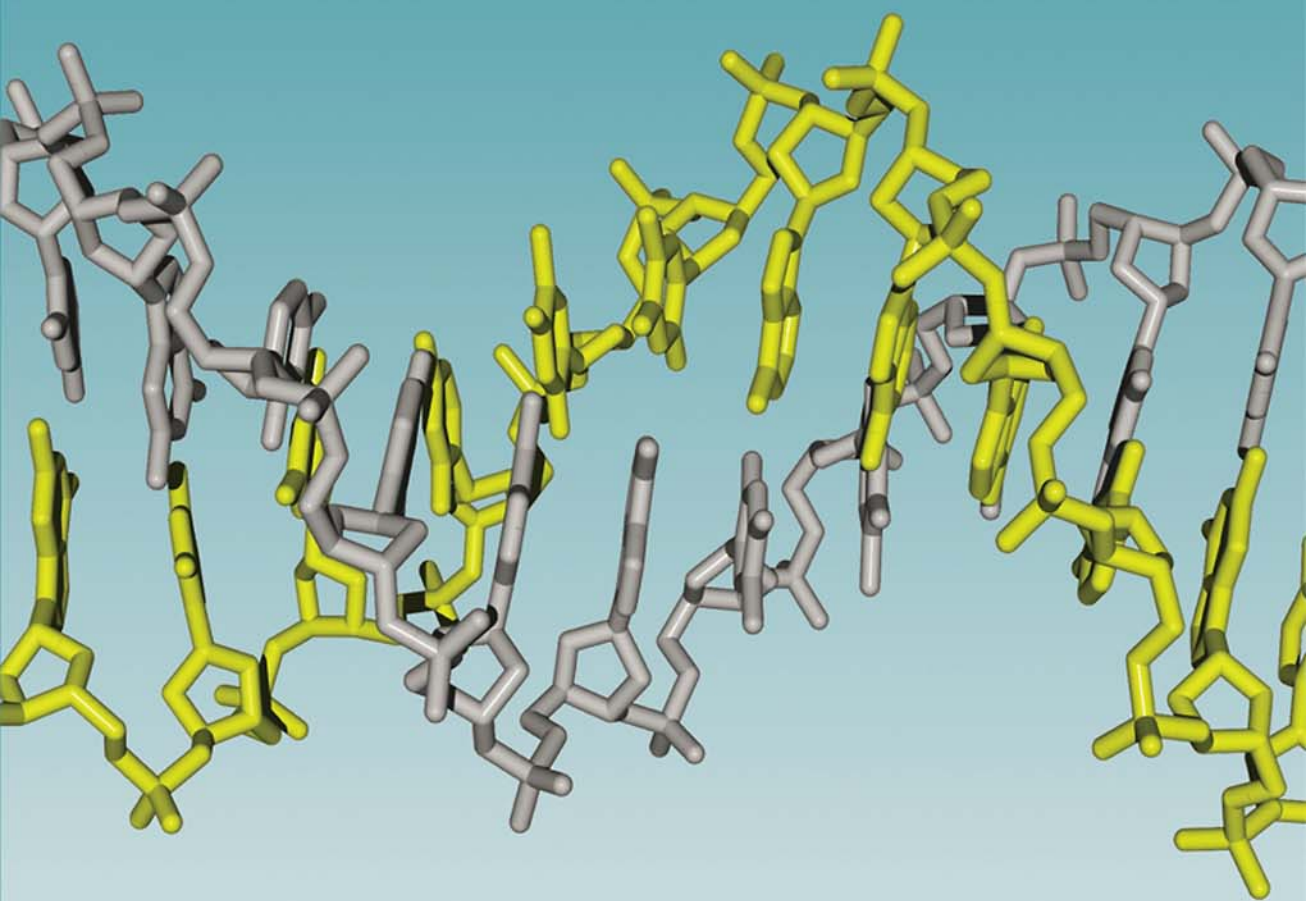


Gene Cloning

Principles and Applications



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1 Introduction

1.1 The Beginning of Gene Cloning

In November 1973, a five-page paper was published in the prestigious journal *Proceedings of the National Academy of Sciences USA* by Stanley Cohen, Annie Chang, Herb Boyer and Robert Helling from Stanford University in California. The title of the paper was “Construction of biologically functional plasmids *in vitro*”, and it described for the first time the production of an organism into which a DNA molecule had been introduced which consisted of DNA sequences from two different sources, joined together in the test tube. Although this work itself built on an earlier body of research, it may justifiably be seen as the paper which marked the birth of a scientific revolution which has continued to this day.

Great changes in science come about in different ways. Sometimes, they are the result of new concepts that transform our way of looking at things, or give us new insights into areas of knowledge which had previously been obscure. Such a revolution in biology had already occurred in the two decades before Cohen’s paper, with the realization that the fundamental stuff of inheritance is DNA, with the discovery of DNA’s remarkable structure, and with the unscrambling of the genetic code. Other dramatic changes in science have been more technical than conceptual, and are no less important for that. Cohen’s paper describes the first methods for manipulating DNA in ways that began to give the experimenters a measure of control over these molecules, hence enabling manipulation of the genetic properties of the organisms that contain them. Humans have, of course, been selectively breeding organisms for particular traits for millennia: domestication of wild plants for crops was a hugely successful early experiment in genetic engineering. But with the advent of what are commonly called recombinant DNA techniques, the degree to which we can produce predetermined genetic changes with high precision has grown to the point where now it is commonplace to make bacteria or plants that produce human proteins, to tinker with the basic structures of enzymes to

alter their activity or stability, or to pull a single gene from the tens of thousands present in a human chromosome and identify within it a single changed base that may give rise to a crippling genetic disease.

One of the hallmarks of the maturity of a technology is the extent to which it works its way through the system. It begins as the preserve of one or a few specialized laboratories; then, it becomes more widely available and used, and commercial applications begin to appear; ultimately, it makes its way onto undergraduate and even school curricula. Experiments that were once the material of Nobel prizes become the topic of routine practicals. For many years we have run, here in our own school, a practical for second-year undergraduates which is, in essence, not that dissimilar to the breakthrough experiment described by Cohen and his colleagues in 1973. Many hundreds of our undergraduates have become gene cloners by the end of their second year at university, and this is also true in thousands of institutions all over the world, including probably the one where you are studying. In addition many aspects of the biology that are taught are only known and understood today because of the incredible power that recombinant DNA methods give us to answer fundamental questions about the nature of life and the functions of cells and organisms. Many of our graduates go on to use these methods extensively in their own careers as research scientists in academia and industry.

If the basic technique described by Cohen *et al.* was all there was to recombinant DNA methods, our life as academics teaching molecular biology would be very simple. But, in fact, Cohen and his colleagues were in many ways the Wright brothers of the field, and in little more than three decades since they published their paper, we have moved from fragile biplanes to jumbo jets. Today's research laboratories have access to hundreds of different approaches to biological investigation which use aspects of recombinant DNA techniques, and whole industries are founded upon their exploitation. Methods have been introduced and refined at a dizzying pace that shows no obvious sign of relenting. Some – such as the ability to exponentially amplify vanishingly small amounts of DNA in a test tube, to manipulate the germ line of complex multicellular organisms, or to determine the complete sequences of the genomes of many organisms – have been revolutions in their own right. Others represent incremental improvements to basic techniques which have nonetheless transformed complex methods into processes that can be done using off-the-shelf kits, or (increasingly) performed by robots. This presents something of a problem for us as teachers, or rather two problems. The first is that the pace of change is such that it is difficult to know what to include and what to omit from undergraduate courses on the subject, and hard to find text books at the right level that are up to date. The second is that the essentially technical nature of the recombinant DNA revolution means it is important to present the subject in such a way that it is not just a dry list of methods, but

which also conveys a sense of the excitement and insight that these approaches have brought to so many different areas, not only in the research laboratory but also in everyday applications. Hence the book that you are now reading. In it we have tried to present a selection of what we regard as the key concepts and methods that underlie gene cloning, at a level which should be easily understandable to a typical undergraduate in a bioscience or medical subject, and to illustrate these as much as possible with examples drawn from the laboratories of universities and companies around the world. Our aim throughout has been to be as comprehensive as possible both with basic methods and with their more advanced applications, subject to space constraints. Inevitably, we have had to be selective in the material that we have covered, and even during the course of writing the book we have had to go back and revise or add to early material as new methods have been published. But we believe that the major aspects of the subject are all here, presented in a form that you will find easy to understand, and which will interest and enthuse those of you that read and use it.

1.2 How To Use This Book

The layout of the book is quite traditional, with the different chapters dealing with methods and concepts of increasing complexity through the book. Although we have made the individual chapters self-contained, and used extensive cross-referencing between chapters, we expect most people will start at the beginning and work their way through the book as needed according to the course they are studying. Each chapter starts with a list of “learning outcomes” – that is, a list of the things you should be able to do once you have read and understood the material in the chapters. These should help you to assess whether you have understood what the chapter is all about. By way of an introduction to the book we present some information about the way genomes are organized in both prokaryotic and eukaryotic organisms. There follows a group of chapters which present basic details about the enzymes and reactions used in simple gene manipulations, and then go on to talk about how genes are actually cloned and identified. We have gone into the details of how clones of particular genes are found, since our experience has been that this is an area that students often find difficult to understand. The advent of high-throughput genome sequencing and the consequent availability of huge amounts of gene sequence data online means that approaches to gene cloning have changed a lot in recent years, but we feel it is still important for you to understand the “traditional” (i.e. more than 10 years old!) methods, even though the use of gene libraries is becoming less common.

It would be ridiculous in a book of this nature, however, not to give a good deal of weight to the topic of genomics (i.e. all aspects of studying organisms at the whole genome level), since this constitutes one of the more recent revolutions in the methodology of the biosciences. Two chapters

describe how DNA is sequenced and how the large amounts of sequence data deposited in international databases can be mined and analyzed – although we have not gone into this latter area in too much technical detail, since this is a whole new discipline in its own right and requires skills in mathematics and computer programming which are beyond the remit of this book.

We then turn to more applied aspects of recombinant DNA methods, including how cloned genes can be used as the source of large amounts of proteins, and how genes can be manipulated and introduced into higher organisms to produce so-called transgenic organisms, the uses of some of which are described as case studies. We discuss also some of the powerful research uses of these methods, such as deepening our understanding of how genes are regulated in cells, and enabling us to functionally dissect proteins. Finally, we conclude with a chapter which discusses some further applications, mainly in a medical context, to add to those used as illustrations in earlier chapters.

Although most of the text will be self-explanatory, assuming you have a degree of basic knowledge (the things we expect you to already know are listed in the next section), there are some places where particular general concepts seemed to us to be sufficiently important that we have put them in boxes, separate from the rest of the text.

One thing that you will notice in the book is the use of large numbers of examples, based on genuine experiments and published results, to illustrate the points that we are making. One of the features of molecular biology is that the methods can be applied to all living organisms, and you will find that in some cases, our examples will be based on bacterial systems (prokaryotes), and on others they will refer to eukaryotes, ranging from single-celled organisms such as yeast all the way to humans. We (the authors) have research and teaching experience both with prokaryotes and eukaryotes, and it has been our experience that it is often best to discuss the simpler prokaryotic systems first, to introduce basic concepts, before going on to talk about the more complex eukaryotes. At the end of each chapter, we have included references for the papers which are referred to in the case studies discussed in that chapter. In most cases, these are available online; if not, they should be in your institution's library. Reading these papers should add to your understanding of the methods and their applications discussed in this book. Some of the papers are quite straightforward, while others are complex and may be tricky to follow in places. However, learning to read the scientific literature is an essential part of any undergraduate degree, and we encourage you to read as many of these papers as you are able. A key feature of the book is the questions that are included in the text of each chapter. Some of these are simply designed to make sure that you have taken in what you have just read, by (for example) setting simple problems based on the previous sections. Others do require a bit of extra

thought, or the bringing together of several different topics. As it is only by trying to answer questions on it that you can really tell how good your understanding of a subject is, we encourage you to persevere with these questions, even if they appear difficult at first, before turning to our answers at the end of each chapter.

1.3 What You Need To Know Before You Read This Book

In writing this book, we have tried to pitch it roughly at a level that would be understandable by undergraduates in the UK in their second year, although some of the more advanced material would perhaps be left until the following final year, and these are the levels at which we have experience of teaching these topics. It is important to be clear, therefore, that this is not a textbook about fundamental concepts in genetics, cell biology, or biochemistry, and it is assumed that you will already know these before you start. We take it that anyone studying this book will be familiar with:

- The structure of DNA
- The nature of the genetic code
- The way in which information flows from DNA via RNA to proteins, and the basic nature of the mechanisms (transcription and translation) by which this happens
- The nature of proteins, including the way in which their structure determines their function, and their different roles in cells
- Basic cell biology of both prokaryotes and eukaryotes

If these are not areas that you are familiar with, then much of the material in this book will be hard to follow, and it would be better to study a more basic text first before trying to use the current book.

1.4 A Request From the Authors

We have tried very hard to make this book precise, informative, interesting and correct. Some of the material has been tested extensively on undergraduates here in Birmingham or has grown from material that we have been teaching for many years. Other material is relatively new, and has involved us in a great deal of research of our own, reading original papers and talking with people using methods with which we ourselves were not directly familiar. It is inevitable, however, that the book will contain flaws, and we genuinely do want to hear about these so that in the event that future editions are needed, we can incorporate any suggestions which are made by you for the benefit of other readers. If you have comments or corrections to make, do please send them by e-mail to gene_cloning@bham.ac.uk. We look forward to reading your comments, and we hope you find the book a valuable aid to studying the fascinating and important topic of gene cloning.

Further Reading

Construction of biologically functional bacterial plasmids *in vitro*. (1973) Cohen SN, Chang AC, Boyer HW and Helling RB. Proc Natl Acad Sci USA, Volume 70 Pages 3240–3244.

The first paper to describe the production of an organism that contained DNA sequences from two different sources.

2 Genome Organization

Learning outcomes:

By the end of this chapter you will have an understanding of:

- *the genomic organization of prokaryotes and eukaryotes, and in particular the human genome*
 - *the different types of sequence within the eukaryotic genome: coding, non-coding, non-repetitive and repetitive*
 - *the physical characteristics of chromosomes*
 - *why an appreciation of genome organization is important in the context of gene cloning*
-

2.1 Introduction

The genome of an organism can be defined as “the total DNA content of the cell”, and as such it contains all the genetic information required to direct the growth and development of the organism. For all multicellular organisms this growth and development starts from a single cell, the fertilized egg. In the case of humans the egg develops into an adult comprising approximately 10^{12} cells made up from over 200 different cell types.

As you will be aware the gene is the basic unit of biological information. Most genes code for a protein product: the gene is transcribed to RNA and this RNA messenger is then translated to the protein product. In addition to genes which encode proteins, there are many genes which encode stable RNAs such as ribosomal RNA and transfer RNA. The number of genes contained within the genome of an organism ranges from around 500 for the bacterium *Mycoplasma genitalium* to over 50,000, predicted to be present in most plants.

In bacteria the genetic information is normally carried on one circular DNA molecule referred to as the bacterial chromosome, which may be supplemented with several small self-replicating DNA molecules, also known as plasmids. Eukaryotic cells contain several linear chromosomes within the nucleus. Human cells, for example, contain 23 pairs of chromosomes. In addition to the DNA present in the nucleus, mitochondria and chloroplasts

contain DNA that encodes a fraction of the functions of these organelles. For multicellular organisms the genome content is identical for all cells with only a few exceptions (such as red blood cells, which contain no nuclei and hence no nuclear DNA).

Although the genome and gene content of an organism is necessary for the development and survival of that organism, it is not sufficient. There are important proteins in the fertilized egg whose function is to control how these genes are used. Because of this no free-living organism could be created from its DNA alone. All cells present on Earth today have arisen from pre-existing cells. Genetic engineering cannot lead to the generation of novel organisms from basic components, genetic engineering can only modify the genetic make-up of pre-existing cells by adding or removing functions from the organism's genome.

In order to understand how to manipulate DNA it is important to understand the way it is organized in different organisms. In this chapter we will discuss the main features of the genome of higher eukaryotic organisms using the human genome as our primary example. We will also discuss bacterial and viral genomes so as to understand how they differ from those of eukaryotic organisms.

2.2 The C-value Paradox

The C-value is a measure of genome size, typically expressed in base pairs of DNA per haploid genome. The use of the term haploid genome refers to a single copy of all the genetic information present in the nucleus. Diploid nuclei of organisms produced sexually will of course contain two complete,

Table 2.1 Characteristics of the genomes of example organisms

Organism	Genome Size (bp)	Chromosome Number (n)	Predicted Number of Genes
<i>Mycoplasma genitalium</i>	580,000	1	500
<i>Escherichia coli</i> K12	4,639,000	1	4,500
<i>Saccharomyces cerevisiae</i> (yeast)	12,069,000	16	6,000
<i>Caenorhabditis elegans</i> (worm)	97,000,000	6	20,000
<i>Drosophila melanogaster</i> (fly)	137,000,000	6	15,000
<i>Oryza sativa</i> (rice)	420,000,000	12	40,000
<i>Arabidopsis thaliana</i> (weed)	115,000,000	5	28,000
<i>Fugu rubripes</i> (pufferfish)	390,000,000	22	25,000
Mouse	2,500,000,000	20	25,000
Humans	3,300,000,000	23	25,000

and not quite identical, copies of a haploid genome, each derived from one of the parents. Table 2.1 gives the C-value for a range of different organisms. One surprising outcome of analyzing the C-value from different organisms is the so-called “C-value paradox” which refers to the fact that genome sizes (and hence the C-values) do not always correlate with genetic and/or morphological complexity. The C-value paradox states that the organism with the largest genome is not necessarily the most complex and that genome size cannot be used as a predictor of genetic or morphological complexity. For example humans and mice, in common with most other mammals, have a genome size of around 3 billion base pairs (3×10^9 bp). However the unicellular protozoan *Amoeba dubia* has a genome size of over 600 billion base pairs (6×10^{11} bp) about 200 times as big. The C-value paradox means that organisms with similar complexity may have very different genome sizes and conversely organisms with similar C-values may not be equally complex. The main exception to the C-value paradox is found in the prokaryotic kingdom, where genome size is a good predictor of metabolic complexity, which in turn often relates to the range of different niches within which a particular bacterium can survive.

In prokaryotic organisms like bacteria, genes are packed tightly together with very little non-coding DNA being present, although all higher eukaryotes contain a large amount of repetitive non-coding DNA. The presence of varying amounts of this non-coding DNA in different eukaryotic organisms explains how relatively simple organisms can have more DNA in their genomes than more complex ones. Remarkably, genes which encode proteins are present as oases in a desert of non-coding “junk” sequences. In the human genome, for example, there is on average only one gene for every 100 kb of sequence.

Q2.1. If the genomes of higher organisms were similar to bacteria, how many genes could be encoded by the human genome? Assume that in bacteria genes are about 1 kb in length.

2.3 The Human Genome

The human haploid nuclear genome contains 3×10^9 base pairs, a vast amount of DNA. To put this in context: if you were to begin reading the human genome sequence at 1 base per second and continue for 24 hours a day, 7 days a week, it would take you approximately 100 years to finish. This DNA is organized into linear segments, the chromosomes, which vary in length from 47×10^6 bp to 246×10^6 bp. There are two sex chromosomes (the X and Y), which determine the gender of the person carrying them. The other 22 chromosomes are called autosomes. Somatic cells in diploid eukaryotes contain pairs of each autosome plus two sex chromosomes. The pairs are not perfect pairs, however, in that they are not identical in

DNA sequence, as one originates from the father, and one from the mother, of the person in question. The number of chromosomes quoted for an organism is usually the haploid number (23 for humans). Somatic human cells therefore contain approximately 6×10^9 bp of DNA – two complete (and non-identical) copies of the genome. A recent milestone in science has been the sequencing of the entire human genome.

In addition to the nuclear genome, mitochondria also contain DNA: in humans, the mitochondrial genome is about 17×10^5 bp in length. As each of the 800 mitochondria per cell contains 10 copies of the mitochondrial genome, this is in fact the most abundant DNA molecule in the cell. The abundance of mitochondrial DNA has been exploited in the analysis of ancient DNA (aDNA), and in forensic investigations where limiting starting material can hinder genetic analysis. Mitochondria are maternally inherited, coming almost exclusively from the egg. The maternal inheritance of mitochondria can also be exploited in both population genetics studies and in forensic analysis.

How is the genome organized? As pointed out above, much of the genome in higher eukaryotes is non-coding. Initial analysis has indicated that less than 2% of the human genome actually encodes protein. The remaining non-coding DNA is often referred to as junk DNA and in most cases probably serves no purpose. However, this junk DNA, some of which has arisen due to DNA duplication, has allowed the evolution of new genes and hence the generation of genetic variation. Recent comparison of dog, mouse and human whole genome sequences has shown a surprising degree of sequence conservation in regions of non-coding DNA. Approximately 5% of the mammalian genome is under selective pressure, i.e. 5% of their genome sequences have not changed significantly since mice, dogs and humans diverged from a common ancestor; this suggests that at least 3% of the non-coding DNA is conserved. The conserved non-coding DNA will include regulatory sequences required to control gene expression to ensure that the correct proteins are made at the right time in the appropriate cells. However, some of the conserved sequences may have functions that we do not yet fully understand. The study of this junk DNA will play an important role in understanding how modern genomes have evolved. In addition, as we will see in Section 13.2, this non-coding DNA has been exploited in forensic and population studies. Repetitive non-coding DNA also presents many problems for genome sequencing projects; this will be discussed in Chapter 7.

Non-coding DNA

Non-coding DNA can be grouped into four main categories, namely: (1) introns; (2) simple sequence repeats; (3) interspersed transposon-derived repeats; and (4) non-repeat, non-coding DNA. All these are discussed below. In addition there are also pseudogenes and segmental duplication,

both of which have arisen through the duplication of regions of the genome during evolution. A duplicated region can either be on the same or a separate chromosome.

Introns

Most eukaryotic genes are made up of stretches of DNA that code for amino acids, interrupted by non-coding sequences. The coding and non-coding sections are called exons and introns respectively. An example of eukaryotic gene structure is shown in Figure 2.1, where the gene indicated has three exons separated by two introns. Eukaryotic RNA polymerases transcribe both the exons and introns to yield an RNA molecule called the primary transcript; it requires further processing before it can be used as a template for translation (Figure 2.2). This processing includes removal of the introns in a process called splicing, which edits the exons into a contiguous coding sequence that can be translated by the ribosome. Introns make up 24–28% of the human genome, and the “average” human gene is encoded by nine small exons of 145 bp each separated by large introns of over 3 kb each.

The existence of introns has many implications for gene cloning and analysis. Most genetic engineering requires the manipulation of a sequence that is devoid of introns, i.e. is a single contiguous coding sequence. As you can see from Figure 2.2a, the mature messenger RNA contains an uninterrupted copy of the protein coding sequence. Therefore the easiest way to obtain a contiguous coding sequence is to isolate messenger RNA and to then create a DNA copy. A DNA copy of messenger RNA is called cDNA, and the way in which this is produced and used will be discussed in Section 4.18. Another issue that is raised by introns is the problem of identifying genes within whole genome sequences. This is made particularly difficult because of the fact that small exons are separated by very large introns and predicting exon-intron boundaries is a difficult challenge. The problem of identifying the coding sequences within genomic

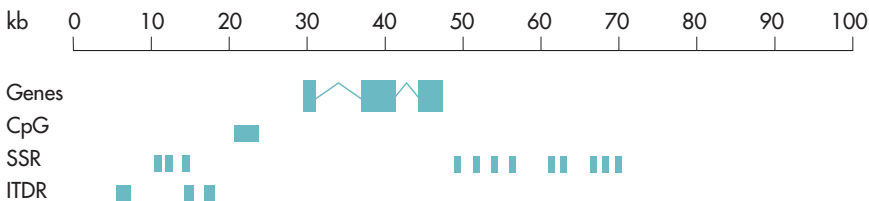


Figure 2.1 Schematic representation of 100 kb of a human chromosome. The four rows represent different classes of sequence found within the genome. This region of the genome contains only one gene, the exons are shown as blocks and the introns as lines. The position of simple sequence repeats (SSRs), interspersed transposon-derived repeats (ITDRs) and CpG islands (CpG) are indicated by the blocks.

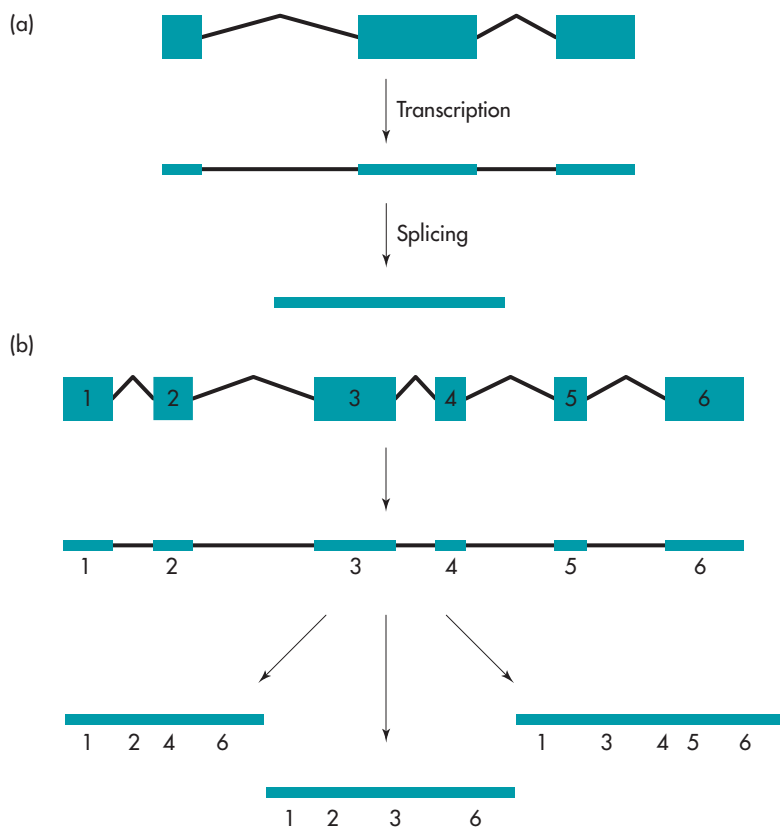


Figure 2.2 Eukaryotic genes contain introns. a) Schematic representation of a gene. Exons and introns are shown as blocks and lines respectively. RNA polymerase transcribes both the exons and the introns, the introns are then edited out in a process called splicing to generate the mRNA molecule that is exported to the cytoplasm and translated. b) A schematic representation of a gene with six exons. Three splice variants are shown.

DNA is therefore generally resolved by analysing cDNA in parallel with genomic DNA; this will be discussed in Section 8.3.

An added complexity of human genome biology is the prediction that at least 35% of human genes are alternatively spliced. This is a much higher level than for any other organism for which genomic information is available. Alternative splicing is shown in Figure 2.2b, which depicts a hypothetical gene that contains six exons. In this example the six exons could be spliced in three different ways leading to three different proteins. It should be noted that the three proteins will share common peptide sequences, i.e. all three have the same N-terminal and C-terminal sequences. Alternative splicing may actually explain how a relatively small number of genes (20–25,000) can code for an organism as complex as a human. Alternative splicing generates the potential for over 100,000 different proteins, so the

relative complexity of the human proteome (the catalog of human proteins) may be due to alternative splicing rather than a high gene number. This impacts on our understanding of development, health and disease. Methods used to study the proteome (total protein profile) and transcriptome (total mRNA profile) of cells will also be discussed in Section 11.6.

Q2.2. Given a gene with the following structure:

E1	Intron	E2	Intron	E3	Intron	E4
----	--------	----	--------	----	--------	----

How many alternatively spliced variants can be generated? Note alternative splicing results in one or more exons being spliced from the final message; it cannot result in the reordering of exons, i.e. a final message with the exon order E1+E3+E4 can occur, but not E3+E1+E4. You should assume that E1 and E4 are present in all spliced mRNAs.

Simple sequence repeats

A region of the genome where a DNA sequence is repeated many times in tandem is called a simple sequence repeat (SSR). These regions are described as satellite DNA, mini-satellite DNA and micro-satellite DNA depending on the number and length of the repeats (see Box 2.1 for details). SSRs, i.e. satellite, mini-satellite and micro-satellite, make up 3% of the human genome. Most SSRs are not associated with a phenotype as they fall in non-coding DNA, however a subclass of micro-satellites, often referred to as trinucleotide or trimer repeats, are within protein coding genes and as such can give rise to altered phenotypes and, in certain circumstances, disease. The rapid evolution of different dog breeds is thought, in part, to be due to changes within these trimer repeats. One example is the repeat 5'-CAG-3' found within the Huntington (hdh) gene. Most individuals have 10–35 copies of the 5'-CAG-3' repeat within each of the two copies of the hdh gene in the cell. However, if the number of repeats exceeds 35 within either copy of the gene, this leads to Huntington's disease. Several other genetic diseases are due to trinucleotide repeat expansion. It was analysis of mini-satellites that formed the original basis of DNA fingerprinting, whereas modern DNA profiling techniques are based on the analysis of micro-satellite DNA (see Section 13.2).

Interspersed transposon-derived repeats

By far the most abundant repeat sequences found within the human genome are the transposon-derived repeats or interspersed repeat sequences, which account for 45% of the human genome. These are derived from transposons, mobile genetic elements capable of being copied and inserted into new sites in the genome. The ability of transposons to spread through the genome is thought to have given rise to an

Box 2.1 Simple Sequence Repeats (SSRs)

Simple sequence repeats (SSRs) are found in all eukaryotes, although their relative abundance does vary between organisms. When genomic DNA is isolated by centrifugation on a cesium chloride gradient it forms a discrete band although smaller satellite bands are often visible. These contain simple sequence repeat DNA which is also known as satellite DNA. SSRs can be classified into three categories, satellite DNA, mini-satellite DNA and micro-satellite DNA.

Satellite DNA sequences are typically greater than 100 kb in length and composed of repeats five to 200 bp in length. An example is the alphoid DNA found in centromeric DNA, in which a 171 bp sequence is repeated many times, resulting in a region of around 500 kb.

Mini-satellite sequences are generally less than 20 kb of DNA composed of repeats less than 25 bp in length. An example of a mini-satellite sequence is one found near the insulin gene, which has the sequence 5'-ACAGGGGTGTGGGG-3'. The number of repeats of the mini-satellite sequence at each locus varies between individuals and they are therefore also referred to as variable number tandem repeats or VNTRs. It was analysis of these VNTRs that formed the original basis of DNA profiling.

Micro-satellite DNA is normally less than 150 bp in length and composed of 5 bp repeat sequences. Micro-satellites are also known as simple tandem repeats (STRs). An example is the D18S51 micro-satellite, which has the sequence 5'-AGAA-3'. Most individuals will have between seven and 27 copies of D18S51 on each copy of chromosome 18. Analysis of these simple tandem repeats has formed the basis of modern DNA profiling techniques (Section 13.2).

accumulation of these sequences in eukaryotic genomes over evolutionary time. Similarly, over time these transposons have been subject to mutations and deletions, resulting in most cases in the loss of the ability to transpose. There are four classes of transposable elements, namely: LTR retroposons (retrovirus-like elements), long interspersed elements (LINEs), short interspersed elements (SINEs) and DNA transposons; a more detailed discussion of interspersed transposon-derived repeats can be found in Box 2.2.

In addition to the multiple repeat elements mentioned above, approximately 5% of the genome appears to be a duplication of another segment of the genome. There are several mechanisms that, during evolution, result in sections of the genome being copied and pasted into the same or different location on the chromosome. Often the duplicated segment will have

Box 2.2 Interspersed Transposon-Derived Repeats

Interspersed transposon-derived repeats are found in all eukaryotes. They are typically the most abundant repeat sequences found within the genome and account for 45% of the human genome. There are four classes of transposable elements, LTR retroposons (retrovirus-like elements), long interspersed elements (LINEs), short interspersed elements (SINEs) and DNA transposons.

LTR retroposons contain long terminal repeats (LTR elements) and are relics of retroviruses. They contain the necessary signals to allow transposition within the genome, a process that requires transcription of the retroposon element followed by reverse transcription to form a DNA copy which then integrates at a random site within the genome. LTR retroposons account for approximately 8% of the human genome.

Long interspersed elements (LINEs) are retroelements, 6–8 kb in length, that do not have an LTR sequence. The LINE sequence encodes two proteins that are involved in transposition of the sequence within the genome. The inefficient mechanism used during transposition often results in truncated forms of the LINEs being transposed within the genome. LINE elements make up approximately 20% of the human genome.

Short interspersed elements (SINEs) are short DNA sequences, 100–300 bp in length, which are not capable of transposition on their own but can do so using functions provided by LINE elements. The Alu element, a 300 bp sequence, is the most abundant SINE with more than 1,000,000 copies present within the human genome representing 10% of the genome. In total, SINEs make up approximately 13% of the human genome.

Active DNA transposons are 2–3 kb in length and contain a transposase gene flanked by terminal repeat sequences; inactive transposons contain the inverted repeats but have lost the transposase gene. DNA transposons make up 3% of the human genome.

LTR retroposons, LINEs and SINEs all transpose via an RNA intermediate, whereas DNA transposons spread through the genome via a copy and paste mechanism.

contained functional genes which therefore leads to redundancy. The duplicated gene or genes can pick up mutations that either lead to change or loss of function. So gene duplication is one way in which genes with new functions can occur and is responsible for the evolution of gene families. Loss of function will result in a pseudogene, i.e. a sequence on the genome that looks like a gene but no longer gives rise to a functional protein.

Non-coding, non-repeat DNA

SSRs and interspersed transposon-derived repeats account for 50% of our genome, while the coding DNA makes up only 1–1.4% of the genome. This leaves approximately 50% of the genome made up of “unique” non-coding DNA, half of which will be present in introns. A small percentage of this non-coding DNA will contain regulatory sequences such as enhancers and promoters that are required for the control of gene transcription (see Section 11.1). It is highly unlikely that the remaining 25% arose from segmental duplication and therefore this must, at some point, have been derived from ancient transposable elements that have changed during evolution such that they are no longer recognized as transposable elements.

The impact of non-coding DNA on genome analysis

What is the relevance of interspersed repeat and simple sequence elements to gene cloning? These elements provide a fundamental problem for assembly of whole genome sequences. All current genome sequencing protocols, at some point, involve generating a random set of overlapping fragments of only a few kilobases in length (Chapter 7). Once sequenced, the fragments need to be assembled into the correct order to yield the finished genome sequence. Clearly, if the genome is littered with repeat sequences, many of which are several kilobases in length, this can lead to mis-assembly of the finished sequence. Segmental duplication gives rise to gene families, which means that it is important to differentiate between a gene that you wish to study and the similar genes within the genome. Pseudogenes must also be taken into account when trying to identify “functional” genes and when predicting the gene content of a particular genome.

The genes and other functionally important sequences

The coding and other functionally important DNA makes up only a small fraction of the human genome sequence. Because of the difficulty in analysing the genome sequence and predicting precisely where all the genes occur, we do not have a complete catalog of all human genes. This is exemplified by the fact that initial analysis of the human genome sequence by the public genome consortium predicted the existence of 30,000 genes, whereas Celera Genomics predicted the existence of 40,000 genes. Although the two groups both identified the 10,000 previously known genes, there was little overlap in the predicted novel genes within their data sets. With the publication of the “finished” human genome sequence in 2004 the number of predicted genes has fallen further. The best estimate for the number of human genes is still a figure of 20,000 to 25,000. Genes are non-randomly scattered around the genome, with gene rich and gene poor regions. Not all chromosomes have the same gene density, as you can see from Table 2.2 which summarizes data from the public genome consortium.

Table 2.2 Human chromosome size and gene content

Chromosome	Size (bp)	Number of Genes
1	246,000,000	2800
2	243,000,000	1800
3	200,000,000	1500
4	191,000,000	1100
5	181,000,000	1200
6	171,000,000	1350
7	159,000,000	1300
8	146,000,000	950
9	138,000,000	1050
10	135,000,000	1000
11	134,000,000	1600
12	132,000,000	1300
13	114,000,000	500
14	106,000,000	900
15	100,000,000	950
16	89,000,000	1050
17	79,000,000	1400
18	76,000,000	400
19	64,000,000	1550
20	62,000,000	750
21	47,000,000	300
22	50,000,000	600
X	155,000,000	1100
Y	58,000,000	130

Figures given are as predicted by the Ensembl project (www.ensembl.org). The gene number represents genes that have been identified or predicted as of November 2004.

Q2.3. Using the data in Table 2.2, determine the gene density for the human chromosomes 13, 19 and the Y chromosome in terms of genes per million base pairs. Which is the most gene-rich and which is the most gene-poor chromosome?

As genome sequencing projects reveal the sequence of other mammalian genomes, it is becoming clear that human developmental and intellectual complexity is not due to increased gene number. All vertebrates carry

roughly the same number of genes and it is possible that flowering plants contain more genes than humans. The mouse genome contains an almost identical set of genes to ours; almost every human gene has a mouse equivalent. So what makes us different from a mouse? The key difference between organisms is how the genetic information is utilized, in particular when and where the genes are expressed. It appears that the DNA itself is less important than what you do with it. If we wish to understand the genetic aspects of human development, health and disease, we need to be able to monitor, on a genomic scale, both the timing and level of gene expression of all genes, as well as the generation of protein diversity due to alternative splicing of mRNAs. These topics will be discussed in Section 11.6.

Variation between individuals: single nucleotide polymorphisms

We all have copies of the same genes, so the question can be asked: What generates the diversity seen within the human population? What makes you different from your siblings and your friends? Diversity arises from the fact that many genes are polymorphic, i.e. there are many different versions of the same genes within any population. In most cases, this variation is due to base changes at a single position. Such variations are called single nucleotide polymorphisms (or SNPs, pronounced “snips”). It is important to differentiate between rare mutations and SNPs. If you were to sequence the genome of tens of thousands of individuals you would probably find very rare mutations at virtually every position within the genome. SNPs, however, are defined as single bases within the genome which differ in at least 1% of the population, so they occur at much higher frequencies than rare mutations.

SNPs are distributed non-randomly throughout the genome, occurring at a frequency of once every 1000 bases. There are two types of SNPs found within the protein coding region of genes. First are those that cause no change in the amino acid sequence of a protein. These SNPs change a base but, due to the degeneracy of the genetic code, the amino acid encoded by the codon of which this base is a part is not altered. These are therefore examples of silent mutations. The second type of SNP does alter the amino acid encoded and may therefore alter the property of the protein. SNPs do not have to occur in coding regions; they can, for example, also occur within promoter regions (sequences that control if, when and to what level a gene is expressed). And, in fact, most SNPs will occur within non-coding regions of the genome, and because of this it has been predicted that less than 1% of SNPs will have a significant phenotypic effect. These SNPs that do have an effect on the phenotype are responsible for almost all genetic variation between members of a species, such as variation in response to drug treatment and variation in susceptibility to, or protection from, all kinds of diseases (Box 13.2).

Even the 99% of SNPs that are predicted to have no direct impact on phenotype are targets of genome biologists since many will be positioned

close to other genes of interest, and they will therefore be linked to these genes. SNPs are also useful markers in population genetics and evolutionary studies. Because of this, SNP study has become a major focus of genome analysis.

Q2.4. Below is a section of a hypothetical gene with the sequence of the encoded protein. Two SNPs are marked that are found in this sequence. Would both SNPs result in an altered protein sequence? Would you expect two SNPs to be this close together?

GGGCTAGCCAACAGGCCAAGGATTGGGAGGAGAGACTAAACATGGTGGGTGACTGTGAT
 GlyLeuAlaAsnArgProArgIleTrpGluGluArgLeuAsnMetValGlyAspCysAsp

2.4 Genomes of Other Eukaryotes

In addition to the human genome, the genomes of many other organisms have been sequenced, including those of a number of model organisms (Box 2.3). With some exceptions, most eukaryotes have a similar genome organization to humans, in that they contain a vast amount of non-coding DNA that contains simple sequence repeats and transposon-derived repeats. A degree of relationship between the complexity of the organism and gene content is observed, so that, for instance, the yeast *Saccharomyces cerevisiae*, the fruit fly, *Drosophila melanogaster* and the thale cress *Arabidopsis thaliana* are predicted to contain around 6000, 15,000 and 25,000 genes, respectively – increasing as the complexity of the organism increases. However, as already pointed out above, this relationship does not always hold true and genome size, or C-value, is not always well correlated with the complexity of the organism.

In eukaryotic organisms the genome is organized into a number of linear chromosomes. The number of chromosomes is essentially arbitrary and sometimes even quite closely related organisms can have different chromosome numbers, with no obvious evolutionary reason or biological consequences. This is classically exemplified by the small muntjac deer, which has a genome almost as large as the human genome. The Chinese subspecies contains 23 pairs of chromosomes, whereas the Indian subspecies contains just three pairs. Both subspecies contain the same genes but in the Indian subspecies the 23 chromosomes present in its Chinese relative have become joined into three very large chromosomes. To date it appears that the Indian muntjac not only contains the smallest number of chromosomes observed in a mammalian species but also the largest chromosome, with one of its chromosomes containing more than 1 billion base pairs of DNA.

The biological significance, if any, of the way in which the genome is organized into different numbers of chromosomes in different species is

Box 2.3 Model Organisms

Throughout the history of biology many experimental scientists have chosen to work on a limited number of “model” organisms that range from bacteria to the chimpanzee. Model organisms are useful to scientists because they provide a standard within each area of research, and by concentrating research on these organisms a large body of information is accumulated. Because fundamental biological processes such as metabolism, development and genetics are conserved by evolution it is possible to extrapolate findings based on studies of relatively primitive model organisms such as the fruit fly or the nematode worm to other more complex organisms such as humans.

The main model organisms tend to be organisms which are easy to grow and to manipulate and for all of the examples listed below extremely powerful genetic techniques have been developed and the genomes have been sequenced. Many experiments discussed in the following chapters have been completed using one or other of these model organisms.

- The bacterium *Escherichia coli*
E. coli is very easy to grow and has a very short doubling time. Virtually all cloning experiments use *Escherichia coli* at some point.
- The yeast *Saccharomyces cerevisiae*
This single-celled organism is the most primitive model eukaryotic organism. It is very easy to grow and to manipulate.
- The weed *Arabidopsis thaliana*
This small plant has a very short life cycle and is easy to grow. It has a relatively small genome for a plant.
- The fruit fly *Drosophila melanogaster*
This organism is easy to grow in large numbers and over many years of research a large number of mutants have been identified and characterized.
- The nematode worm *Caenorhabditis elegans*
These are easy to grow in large numbers and, because they are transparent, every cell in the body can be viewed in live organisms. There is a strictly defined number of cells in the mature organism and the development pathways are well characterized.
- The zebra fish *Danio rerio*
This small fish is a relatively new model organism, it has a transparent embryo and is easy to grow in large numbers. As a vertebrate it is used as a model for some aspects of human biology as it is easier to breed than mice and rats.
- The mouse *Mus musculus*, and the rat *Rattus norvegicus*
These mammals have essentially the same gene content as humans. Powerful genetic techniques have been developed for their manipulation, including, in the case of the mouse, the ability to “knock out”

individual genes. However, these animals are relatively slow to breed and expensive to keep.

- The chimpanzee

This is our closest relative and has been used for experiments which would not be ethical if performed on humans. However, chimpanzees are not cheap to maintain or easy to breed and are not used for routine purposes.

not known. Similar organisms have similar gene sets but these can be arranged differently at the chromosome level. During evolution this genetic information has been moved around, but it does appear to have been reorganized in “blocks”. For example, two chromosomes which are separate in one species may have been joined (fusion) in another, or a single chromosome from one species may have become two in another (fission), but the overall gene order is essentially the same. When the orders of genes are rearranged in closely related organisms, it is often by large inversions or similar events such that the order of genes within the inversion is unchanged but they are reversed with respect to those around them. Therefore if you look at a particular gene homologue in the genomes of related organisms, for instance within the mammals, it is often found with the same neighboring genes. This phenomenon is called synteny. In most cases not only are the genes found together but also they are arranged in the same order, a phenomenon termed colinearity.

Because of these phenomena, people have begun to ask the following question: how many rearrangements, such as large inversions, fusions, fissions or translocations (where a chunk of DNA is moved from one chromosome to another), would it take to get from the genome of one species to that of a related species? The answer is often surprisingly few. For example, it is possible to take the human genome, fragment it into approximately 200 pieces containing at least two genes and then rearrange these 200 pieces to “create” the mouse genome. (This “created” mouse genome is not identical to the actual mouse genome, as sequences of individual genes differ between mice and men, but the order of genes is essentially the same.) The relationship between human chromosome 11 and several mouse chromosomes is shown in Figure 2.3. Thus there are at least two levels of variation to consider when comparing genomes: variation at the DNA sequence level between individual genes, and variation in the order and arrangement of gene homologues along the chromosomes. Synteny and colinearity can be exploited in DNA analysis, and this will increase in future genome sequencing projects where comparative maps will help with the assembly of the genome sequences; for example, the human sequence has been used to help assemble the mouse sequence.

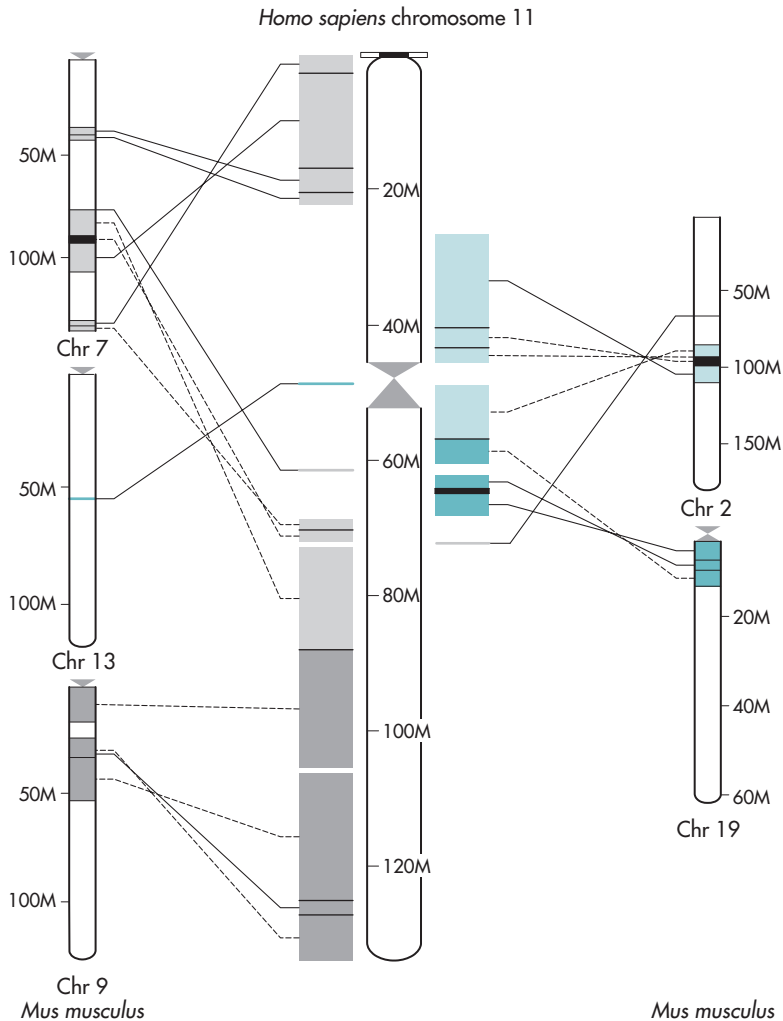


Figure 2.3 Synteny between human and mouse. A representation of the relationship between human chromosome 11 and mouse chromosomes 2, 7, 9, 13 and 19. Shown are the regions of the mouse chromosomes that would need to be rearranged to form human chromosome 11.

You may at this point be asking: since we have sequenced the human genome what is the value of sequencing other genomes, such as that of the mouse? There is still much to learn about the human genome, we still have not identified all of the regions which potentially encode genes, and once a gene has been “discovered” determining its function is another major undertaking. These problems are much more tractable if you take a “comparative biology” approach. Evolutionary pressure has meant that genes and other important sequences, such as control regions, change at a much

slower rate than junk DNA. (To be more accurate, the rate of change at the DNA level is probably roughly constant for all bases, but changes in control and coding regions are much more likely to be deleterious to the organism than changes in the junk DNA and so these changes will be selected against.) The net effect of this will be that changes in the control and coding regions will be less common than those in the junk DNA. Therefore when you compare genomes from different organisms the sequences of interest are highlighted as regions of sequence similarity against the junk DNA background. An example of the power of comparative biology comes from studies on the pufferfish, *Fugu rubripes*. *Fugu* contains the same number of genes as humans but has a relatively compact genome of 365 million base pairs when compared to other vertebrates, i.e. it is one eighth the size of the human genome. The compact size is due to the presence of very little repetitive DNA and the small size of its introns. *Fugu* contains one gene for every 10 kb of genomic sequence. Comparison of the *Fugu* and human sequences has shown that even in these two vertebrates that are separated by 450 million years of evolution, there is some synteny. More importantly, comparative analysis has led to the identification of 900 novel putative human genes.

By comparing the human genome sequence with the genome sequence of more distantly related animals including other mammals, we can catalog important sequences, including genes and regulatory regions, because they have been conserved throughout evolution. In contrast, by comparing the human genome with the recently sequenced chimpanzee genome it is hoped that we will be able to identify the important differences that make us human. The human and chimpanzee genomes that have been sequenced only differ by 1.23%. However, some of the differences are within SNPs that are present in both genomes so at these positions some human and chimpanzee individuals will have the same sequence; the “real” difference between the human and chimpanzee genomes may be less than 1.06%. The fact that most human genes have homologues in all other mammals means that we can use animal models for most human genetic diseases to develop treatments that can then be used in human medicine and, in many cases, veterinary medicine.

As with animals, the size of plant genomes varies considerably: from around 50 megabases to over 100 gigabases. Plants thus exemplify the C-value paradox particularly well. For example, the wheat genome is over five times the size of the human genome. The main reason for the differences in genome size is related to the number of copies of repeat sequences within the genome: organisms such as wheat that have large genome sizes tend to have genomes littered with genome-wide repeats. For example, retrotransposons make up over 70% of the maize genome. The regular occurrence of transposons within plant genomes has been exploited in mapping studies. Another aspect of plant genetics is the phenomenon of

polyploidy. Many plant species, especially flowering plants, have more than two sets of chromosomes per cell. This means that many plants have hundreds of, and in some cases over a thousand, chromosomes per cell. This has an impact on many aspects of gene cloning and analysis. Genome sequencing projects have to sequence the complete genome including all the “junk” and then identify the genes and “useful” DNA. As we will see in Section 4.20, an alternative to sequencing the complete genome is to sequence only the transcribed DNA in the form of expressed sequenced tags (ESTs).

In addition to the mitochondrial genome present in all eukaryotes, plants carry genetic information within their chloroplasts. The circular chloroplast genome encodes approximately 100 genes.

Most plants contain 40,000 to 50,000 genes and it is likely that there is little difference in gene function between different plants, i.e. plants contain virtually all the same genes. As is seen in animals, it appears that related plant species not only share the same genes but that these genes are arranged in the same order on the DNA: there is synteny and, in most cases, colinearity.

Q2.5. What accounts for the difference in size of the genomes of different eukaryotes?

2.5 Bacterial Genomes

Most bacterial genomes are single circular DNA molecules that are 0.5–10 Mb in size. Many bacterial genomes have been sequenced including that of *Escherichia coli*, which is not only a model organism but is the organism which is by far the most widely used for gene cloning. The genomes of many pathogenic bacteria have been sequenced including *Mycobacterium tuberculosis* (which causes TB), *Treponema pallidum* (which causes syphilis), *Rickettsia prowazekii* (which causes typhus), *Vibrio cholerae* (which causes cholera) and *Yersinia pestis* (which causes plague). Comparative analysis of bacterial genomes is proving to be a powerful tool in enhancing our understanding of genome organization in bacteria.

The size of bacterial genomes varies considerably. However, for bacteria, genome size correlates well with gene number, which in turn correlates with morphological, physiological or metabolic complexity. Bacteria with small genomes encode a small number of genes and tend to be restricted to growth in relatively few specialized niches; they are often parasites. Bacteria with large genomes encode many genes and have much higher metabolic diversity. So, for example, the 580 kb genome of the intracellular human parasite *Mycoplasma genitalium* is thought to encode just 500 genes, whereas the 7000 kb genome of the nitrogen-fixing bacterium *Mesorhizobium loti*, which colonizes leguminous plants, encodes

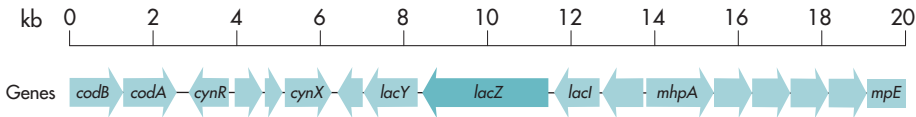


Figure 2.4 Schematic representation of 20 kb of the *Escherichia coli* chromosome. The genes are represented as blocks and the non-coding DNA as lines.

approximately 6800 genes. The average size of a gene in all bacteria is around 1000 bp; therefore, given the examples mentioned above a simple calculation tells you that bacterial genomes are densely packed with genes, i.e. the 500 genes of *M. genitalium* account for 500 kb of its 580 kb genome. The fact that bacterial genomes are packed with genes is clearly demonstrated in Figure 2.4, which shows a schematic representation of 20 kb of the *E. coli* genome (compare this with Figure 2.1). In bacteria there is very little wastage of DNA and little repetitive DNA as seen in higher eukaryotes. Only about 10% of the genome is non-coding and even this 10% in most cases plays a critical role to the cell since it contains the control signals required to co-ordinate patterns of gene expression.

The relatively small genome size of bacterial genomes, together with the fact that they contain very little non-coding or repeat DNA and that they do not contain introns, has made bacterial genomes ideal candidates for whole-genome sequencing projects. This is because, as will be discussed in Chapter 7, a relatively quick and successful strategy for whole-genome sequencing involves sequencing randomly generated overlapping fragments of only a few kilobases. These fragments are then assembled into the correct order to yield the finished genome sequence. As bacterial genomes contain very few repeat sequences assembly is relatively routine. Identification of potential genes within bacterial genomes is also much more reliable, because firstly bacterial genes do not contain introns and secondly the genes are much more closely packed. Potential genes are readily identified by computer analysis (Section 8.2).

2.6 Plasmids

Plasmids are nonessential extrachromosomal elements that control their own replication. They are found mainly in bacteria and in some eukaryotic microbes such as yeast and algae. Plasmids are normally circular double-stranded DNA molecules that range in size from as little as 1 kb to over 100 kb although linear plasmids have been observed in *Borrelia burgdorferi*, the causative agent of Lyme disease. Relative to the size of the bacterial chromosome plasmids are very small. The number of copies of a plasmid in each cell is tightly controlled with a general rule that small plasmids

tend to have a high copy number, sometimes over 100 copies per cell, whereas larger plasmids may be present in one or a few copies per cell. Certain classes of plasmids are capable of horizontal transmission from one bacterium to another. In most cases the recipient bacterium does not have to be the same species as the donor bacterium and can, in fact, be distantly related. By definition, the coding potential of a plasmid is non-essential for the bacterial life cycle but plasmids frequently encode functions that give the bacteria a selective advantage in certain environments, the most infamous example being the class of genes which encode antibiotic resistance. The fact that plasmids can confer resistance to antibiotics and that horizontal transmission can occur between distantly related bacteria has major implications for medicine, since it has allowed rapid spread of antibiotic resistance throughout the microbial world. As you will see in the next chapter, plasmids have been absolutely key to the development of gene cloning.

Q2.6. What are the major differences between the genomes of bacteria and eukaryotes?

2.7 Viral Genomes

For all free-living organisms the genome content is double-stranded DNA. However, viral genomes consist of either RNA or DNA which can be either single- or double-stranded. Since viruses are obligate intracellular parasites they do not need to encode all the functions required by a free-living organism and accordingly they typically have very small genomes. The genome sizes of viruses, which may infect bacteria or eukaryotic cells, vary from 2 kb to 700 kb. It is beyond the scope of this book to discuss the complexities of viral genomes. However, viruses have been exploited throughout the history of gene cloning and as such two key viruses will be discussed below.

Bacteriophage λ is a virus that infects various bacteria including *E. coli*. It has a 48 kb double-stranded DNA genome packaged into an icosahedral head. Its genome is linear. When the λ DNA enters *E. coli* the ends of the linear genome join together to produce a circular genome. Upon infection λ can either follow a lysogenic or lytic pathway. During the lysogenic pathway, the viral genome is integrated into the host genome and is then replicated as part of the host chromosome: no expression of viral structural proteins occurs in this case. If λ enters the lytic pathway (which it can do either directly after infection or by induction from the lysogenic state), proteins required for the formation of progeny virions are made and the viral genome is replicated. Progeny virions are formed, the cell lyses, and about 100 new virions are released. The virus contains a very compact genome, the 48 kb coding for 46 genes. λ has been exploited as a vector for DNA

cloning predominantly because of the efficiency with which it inserts its DNA into *E. coli* (Section 4.10).

Retroviruses are a class of viruses which infect eukaryotic cells that contain a single-stranded RNA genome in a protein capsid surrounded by a membrane envelope. The simplest retroviruses contain an 8 kb genome encoding just three genes, *gag*, *pol* and *env*. The *gag* gene encodes the proteins required to form the viral capsid, the *pol* gene encodes three enzymes, a protease, reverse transcriptase and an integrase, and the *env* gene encodes proteins that are present in the outer membrane. Upon entry into the host cell a double-stranded DNA copy is made of the RNA genome in a process termed reverse transcription, which requires the reverse transcriptase function encoded by the *pol* gene. The integrase then integrates the DNA copy of the viral genome into the host genome at a random position; integration is permanent. The host genes are transcribed and translated, and the immature polypeptide chains are cleaved by the viral protease to yield mature protein products. The exploitation of reverse transcriptase in gene cloning and analysis will be discussed in Chapters 4 and 11.

2.8 GC Content

One characteristic of an organism's genome which we have not considered so far is the DNA base composition; that is, the relative abundance of the bases guanine (G), cytosine (C), adenine (A) and thymine (T). As a result of base pairing there are always equal amounts of G and C, and equal amounts of A and T, and the base composition is usually expressed either as the ratio of (G+C) to (A+T) or as the percentage GC content. The GC content of related species, and of strains within a species, is constant. However, there is considerable variation between different species. The percentage GC content can range from 25–80% in micro-organisms, and 40–60% for higher eukaryotes. In Section 8.2 we will discuss how the GC bias can be used during sequence analysis when trying to identify potential protein coding genes.

During evolution bacteria have often acquired DNA via horizontal transfer and as this DNA may not have originated from the same species, it may have a different percentage GC content to that of the other DNA in the organism. Genome-wide analysis of GC content can thus often highlight regions of DNA that have been acquired via horizontal transfer. In pathogenic bacteria (i.e. bacteria which cause disease) these regions may represent “pathogenicity islands”, the name given to regions of the genome which are distinct from the rest of the genome and which contain some of the genes which render the organism harmful. The human genome on average has a GC content of 40%; however, it does contain GC-rich and GC-poor regions which may have different properties such as gene density and repeat sequence content. In addition, differences in GC content correspond with the occurrence of bands which can be seen on stained

preparations of chromosomes. For example, low GC content corresponds with dark G-bands which are gene poor, whereas high GC content corresponds with the gene-rich interband regions.

Q2.7. When DNA is heated the two strands come apart, in a process termed melting, and the DNA becomes denatured. What effect would the GC content of DNA have on the temperature at which DNA melts, and why?

2.9 Physical Characteristics of Eukaryotic Chromosomes

Chromosomes can be observed by microscopy at metaphase when the DNA becomes condensed prior to cell division. When the metaphase chromosomes of eukaryotes are observed under a microscope it is clear that they are not all the same. Not only do they vary in size, but the position of the centromere is different in, and hence characteristic of, each pair of chromosomes (Figure 2.5). Further detail can be revealed by staining the chromosomes, which produces characteristic banding patterns. Depending on the stain used these bands relate to differences in chromatin structure, such as the degree to which the DNA is condensed.

2.10 Karyotype

As we have already seen, metaphase chromosomes of eukaryotic organisms vary in appearance. Chromosomes such as human chromosome 1, which contain a centromere close to the middle of the chromosome, are referred to as metacentric, whereas in acrocentric chromosomes (e.g. human chromosome 13) the centromere is close to one end of the chromosome. If the centromere is off-centre (e.g. human chromosome 4) the chromosome is called sub-metacentric. Each chromosome has two arms, one either side of the centromere. The short arm is labeled p, for petite, and the long arm q, because it follows p alphabetically.

Metaphase chromosomes can be treated enzymatically, fixed and stained with one of several dyes. A karyotype can then be produced by arranging the chromosome content of the cell in homologous pairs of decreasing size (Figure 2.5). The human male karyotype consists of 46 chromosomes, 22 pairs of autosomes, one X chromosome and one Y chromosome. Historically, the chromosomes have been numbered according to their apparent size, from the largest (chromosome 1) to the smallest (chromosome 22 in humans). This numbering is not always correct; for example, with humans further analysis revealed that chromosome 21 is actually smaller than chromosome 22 and as such is the smallest chromosome. The dye most commonly used is Giemsa stain (often following treatment with trypsin) which produces a distinctive banding pattern comprising alternating dark and light bands, often referred to as G-bands and interbands respectively. Depending on the resolution of the technique

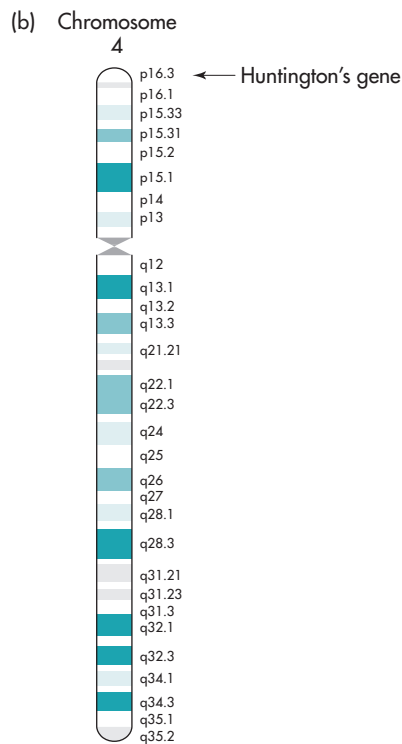
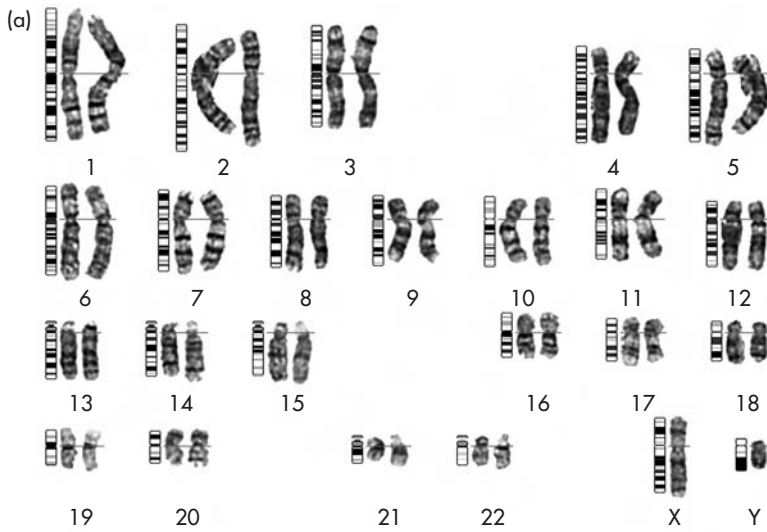


Figure 2.5 The human karyotype and example ideograms. a) The karyotype shows all human chromosomes together with an ideogram indicating the main chromosomal G bands. Note that chromosome 1 is metacentric, chromosome 13 is acrocentric, whereas chromosome 4 is submetacentric (courtesy of Lionel Willatt, Addenbrookes NHS Trust, Cambridge, UK & Digital Scientific, Sheraton House, Castle Park, Cambridge, UK). b) Ideogram representing chromosome 4.

several hundred G-bands and interbands can be visualized for each chromosome. This is often shown as an idealized diagram (ideogram) (Figure 2.5). Bands are numbered consecutively away from the centromere. So, for example, band 4p16, identifies chromosome 4 short arm, region 1, band 6. This low resolution band can be further sub-divided, for example, 4p16.3 refers to sub-division 3 of band 4p16. Often the location of a gene will be described relative to the G-band or interband designation; for example, the gene linked to Huntington's chorea (Huntington's disease) has the chromosomal location 4p16.3 (see Figure 2.5).

Examination of chromosome structure can play an important part in genetic analysis. Changes that are visible to the eye using a light microscope are an indicator of gross changes within the genome. These aberrations can either be inherited from one of the parents or can occur during gametogenesis. Often changes in chromosome structure are not tolerated and hence the fertilized egg containing such changes does not implant or the fetus is miscarried. Changes that are tolerated and lead to fetal development may, however, lead to disease, birth defects or mental retardation. The most common chromosomal change is alteration in chromosome number, referred to as aneuploidy. Monosomy is where there is one missing chromosome, and trisomy refers to a condition where there is one extra chromosome. In humans, for example, in trisomy 21 (Down syndrome) there is an additional copy of chromosome 21. In most cases aneuploidy results in failure of the fertilized egg to implant or, where implantation occurs, spontaneous miscarriage. The only numerical chromosomal aberrations that survive to term in humans are trisomy 21, trisomy 18, trisomy 13, sex chromosome aneuploidy, and triploidy (three complete sets of chromosomes).

In addition to yielding information concerning gross changes in chromosome structure linked to disease, banding gives information about chromatin structure and genome compartmentalization. When the chromosome is stained with Giemsa stain the dark bands (G-bands) have a tendency to be AT-rich and contain relatively few genes whereas the interbands tend to be gene-rich.

Q2.8. How many chromosomes would there be in a human cell displaying (a) monosomy, (b) trisomy, (c) triploidy?

2.11 Euchromatin and Heterochromatin

During mitosis, eukaryotic chromosomes become much shorter due to the DNA and protein in the chromosome taking up a highly structured form, whereupon it is said to be condensed. In eukaryotes, interphase chromosomes are decondensed compared with metaphase chromosomes, but they still contain significant structure. When cells in

interphase are treated with a variety of dyes that stain the nucleus, two distinct classes of DNA are observed: lightly staining regions called euchromatin and densely staining regions called heterochromatin. Euchromatin contains most of the transcriptionally active genes, whereas heterochromatin contains few genes and is composed of repetitive non-coding DNA. Certain regions of the genome are packaged as heterochromatin in all cell types and at all stages of development, and are referred to as constitutive heterochromatin, whereas other regions of the genome may be packaged into heterochromatin in one cell type and euchromatin in others, and are therefore referred to as facultative heterochromatin regions. One method employed by the cell to silence genes permanently is to package the genes in heterochromatin. This has implications for the generation of transgenic organisms since current techniques rely on introducing the gene of interest into the fertilized egg or embryonic stem cells and then looking for transgenic organisms where the gene has been incorporated into the host genome. If the gene inserts in a region of the genome that is destined to be heterochromatin then the gene will be silenced leading to a lack of expression of the transgene and no phenotype for the transgenic organism.

2.12 CpG Islands

A characteristic of mammalian genomes is the occurrence of so-called “CpG islands”. The dinucleotide CpG is unusual because it is under-represented within the genome, occurring at a frequency of less than 1%. Given that the average GC content of the human genome is around 40%, you would expect that the dinucleotide CpG would occur at a frequency of approximately 4%, a figure derived by multiplying the fraction of Cs and Gs within the genome (0.2×0.2). The explanation for this discrepancy is believed to be due to the fact that most CpG dinucleotides are methylated on the cytosine base, i.e. they contain 5-methylcytosine. Spontaneous deamination of 5-methylcytosine or cytosine generates thymine or uracil respectively. Any uracil residues generated by deamination will be repaired back to cytosine by the cell, since uracil should not be present in DNA. However, the product of 5-methylcytosine deamination, thymine, is a normal component of DNA and is therefore not removed. This means that over time methylCpG dinucleotides will slowly mutate to TpG, and so will be lost from the genome. The genome does contain “CpG islands” where the CpG dinucleotide content is approximately 4%, a frequency that matches that predicted from the GC content. These CpG islands contain CpG dinucleotides that normally contain unmethylated cytosines. These CpG islands are important because they occur at the 5' end of genes and, as you will learn in Chapter 8, have played an important role in identifying genes and predicting the number of genes within the genome.

Q2.9. Draw a cartoon of a section of a human chromosome containing a gene. Identify the location of exons, introns and CpG islands. For this gene to be transcribed and translated in a cell would it need to be in euchromatin or heterochromatin?

Much of our understanding of the way in which the genomes of animals, plants and bacteria are organized comes from studies that have only become possible because of the many ways in which we are able to manipulate DNA. We can isolate particularly interesting DNA sequences, sequence them, even change them and look to see what the effect is. It is now an almost routine proposition to sequence the whole of a bacterial genome, and we have the entire genome sequence of an increasing number of plants and animals. Gene cloning and the techniques that allow us to manipulate DNA, have driven developments in genomics, and in turn our increased understanding of genomes informs the way in which we manipulate DNA. In the following five chapters we will look in detail at the basic tools required for the manipulation of DNA.

Questions and Answers

Q2.1. If the genomes of higher organisms were similar to bacteria, how many genes could be encoded by the human genome? Assume that in bacteria genes are about 1 kb in length.

A2.1. *From Table 2.1 we can see that the size of the human genome is 3,300,000,000 bp, long enough to encode 3,300,000 genes of 1 kb. In fact the human genome encodes somewhere between 20,000 and 25,000 genes, most of them bigger than bacterial genes.*

Q2.2. Given a gene with the following structure:

E1	Intron	E2	Intron	E3	Intron	E4
----	--------	----	--------	----	--------	----

How many alternatively spliced variants can be generated? Note alternative splicing results in one or more exons being spliced from the final message; it cannot result in the reordering of exons, i.e. a final message with the exon order E1+E3+E4 can occur, but not E3+E1+E4. You should assume that E1 and E4 are present in all spliced mRNAs.

A2.2. *There are four possible alternatively spliced products: E1+E2+E3+E4, E1+E3+E4, E1+E2+E4, E1+E4.*

Q2.3. Using the data in Table 2.2, determine the gene density for the human chromosomes 13, 19 and the Y chromosome in terms of genes per million base pairs. Which is the most gene rich and which is the most gene poor chromosome?

A2.3. *Chromosome 13 encodes 500 genes and is 114 million base pairs long. The gene density is therefore 4.4 genes per million base pairs. This is the least gene rich of the autosomal chromosomes. The Y chromosome only encodes 130 genes and has a very low gene density of 2.2 genes per million base pairs. Chromosome 19 has the highest gene density at 24.2 genes per million base pairs.*

Q2.4. Below is a section of a hypothetical gene with the sequence of the encoded protein. Two SNPs are marked that are found in this sequence. Would both SNPs result in an altered protein sequence? Would you expect two SNPs to be this close together?

A
A
↑
↑
 GGGCTAGCCAACAGGCCAAGGATTTGGGAGGAGAGACTAAACATGGTGGGTGACTGTGAT
 GlyLeuAlaAsnArgProArgIleTrpGluGluArgLeuAsnMetValGlyAspCysAsp

A2.4. *The first SNP is in the third position of the codon, a change of C to A will not change the amino acid encoded. The second SNP is in the first position of the codon, the change of a T to an A results in a change from cysteine to serine. Although SNPs occur at a frequency of one every 1000 bases, chance can result in two SNPs very close together.*

Q2.5. What accounts for the difference in size of the genomes of different eukaryotes?

A2.5. *Differing amounts of non-coding or “junk” DNA accounts for the variation in the size of the haploid genome, or C-value.*

Q2.6. What are the major differences between the genomes of bacteria and eukaryotes?

A2.6. *Bacterial genomes are small with densely packed genes and very little non-coding DNA.*

Q2.7. When DNA is heated the two strands come apart, in a process termed melting, and the DNA becomes denatured. What effect would the GC content of DNA have on the temperature at which DNA melts, and why?

A2.7. *There are three hydrogen bonds in a GC base pair and only two in an AT base pair, this means that GC base pairs are more stable and hence require a higher temperature to break the hydrogen bonds, as a result DNA with a high GC content has a higher melting temperature than DNA with a high AT content.*

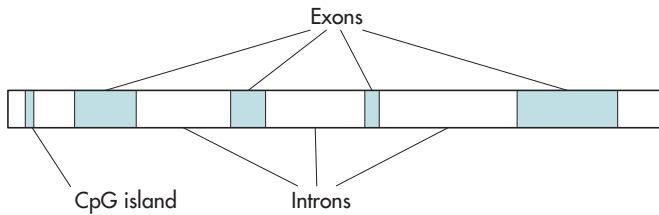
Q2.8. How many chromosomes would there be in a human cell displaying (a) monosomy, (b) trisomy, (c) triploidy?

A2.8. *(a) Monosomy refers to the condition where one chromosome of a pair is missing; the cell would only have 45 chromosomes. (b) Trisomy is where*

there is an extra chromosome resulting in a total of 47. (c) Triploidy refers to a condition where there is a complete set of extra chromosomes, in a human this would mean a total of 69 chromosomes.

Q2.9. Draw a cartoon of a section of a human chromosome containing a gene. Identify the location of exons, introns and CpG islands. For this gene to be transcribed and translated in a cell would it need to be in euchromatin or heterochromatin?

A2.9. *Euchromatin contains most of the transcriptionally active genes.*



Further Reading

Genomes 3 (2006) Brown T, Garland Science, London.

Human Molecular Genetics (2003) Strachan T and Read A, Garland Science, London.

These are two good textbooks that discuss the genome organization and function

Our genome unveiled (2001) Baltimore D, Nature, Volume 409 Pages 814–816.

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The dog has its day (2005) Ellegren H, Nature, Volume 438 Pages 745–746.

A commentary on the publication of the dog genome sequence with links to more in-depth articles

The chimpanzee and us (2005) Li WH and Saunders MA, Nature, Volume 437 Pages 50–51.

A commentary on the publication of the chimpanzee genome sequence with links to more in-depth articles

The mouse that roared (2002) Boguski MS, Nature, Volume 420 Pages 515–516.

A commentary on the publication of the mouse genome sequence with links to more in-depth articles

Single nucleotide polymorphisms ... to a future of genetic medicine (2001) Chakravarti A, Nature, Volume 409 Pages 822–823.

A commentary on the publication of a paper reporting the identification and mapping of SNPs with links to more in-depth articles

3 Key Tools for Gene Cloning

Learning outcomes:

By the end of this chapter you will have an understanding of:

- *what gene cloning is*
 - *the nature of and need for cloning vectors*
 - *what restriction enzymes are and how they can be used to manipulate DNA*
 - *the mode of action of DNA ligase and how it is used to join together different fragments of DNA*
 - *ways of detecting the presence of cloned DNA in a vector*
 - *how PCR can be used to amplify DNA*
-

3.1 Introduction

The discovery of the structure and role of DNA and the unravelling of the genetic code has led over the last 50 years to an explosion in our understanding of organisms and how they work. In the forefront of this revolution came gene cloning. The term “gene cloning” covers a wide range of techniques that make it possible to manipulate DNA in a test tube and also to return it to living organisms where it functions normally. The importance of this technology is that it allows us to isolate any piece of DNA from among the millions of base pairs that make up the genome of an organism. This first step is essential for a whole range of scientific and technological studies, ranging from the study of a gene that is instrumental in causing an inherited disease, to the bioengineering of a strain of yeast that produces a useful pharmaceutical product.

Gene cloning involves taking a piece of DNA from the organism where it naturally occurs and putting it into a cloning host such as the bacterium *Escherichia coli*. It is then possible to study the cloned DNA or produce the protein encoded by the gene. For many applications you may want subsequently to transfer the cloned DNA into another organism, but the initial cloning steps are almost always performed in *E. coli*.

Before looking in detail at how to clone genes, it is important to have an understanding of what the key steps are. DNA is cut into fragments and introduced into a new host, usually *E. coli*, where it is copied. However, you cannot simply introduce fragments of DNA into a cell or organism as they will probably be degraded, and even if it is not it will not be replicated and passed on when the cell divides. To make sure that the piece of cloned DNA is copied and passed on it is necessary to put it into a vector which will ensure that it is copied every time that the cell copies its own DNA and that a copy is passed on to each daughter cell at cell division. This involves cutting the vector and joining in the piece of DNA that you want to clone. This cutting and joining of DNA fragments is done using enzymes. The new molecule that you have thus created is introduced into your host cell by a process called transformation. Once in the host it will be copied and passed on every time the cell divides making many copies or clones of the original fragment.

3.2 Vectors

The most commonly used vectors for gene cloning are plasmids (Section 2.6). Figure 3.1 shows some examples. These are small circular DNA molecules found in many types of bacteria. Plasmids have an “origin of

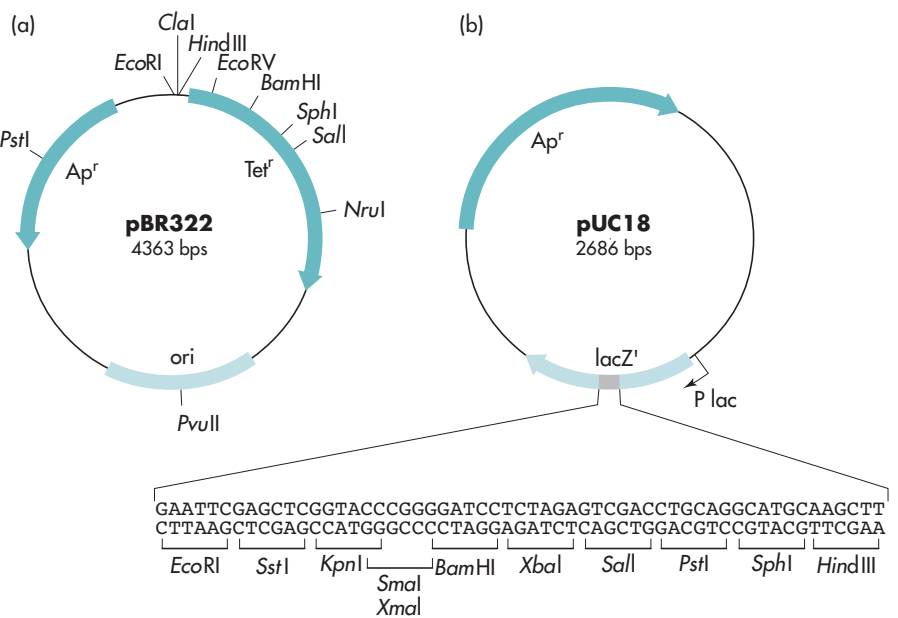


Figure 3.1 a) A map of pBR322 showing the positions of the ampicillin resistance (*Ap^r*) and tetracycline resistance (*Tet^r*) genes, the origin of replication (*ori*) and some of the unique restriction enzyme sites. b) A map of pUC18 showing the position of the ampicillin resistance (*Ap^r*) gene and the *lacZ'* gene; the sequence of the multiple cloning site encoding 10 unique restriction sites is also shown.

replication” which directs the replication of the plasmid and ensures that the cell contains many copies of the plasmid which are distributed between the daughter cells when the cell divides (Box 3.1). The exact number of copies varies according to the particular plasmid. As long as the gene that you have cloned is part of a DNA molecule with an origin of replication, that is, cloned into a plasmid, it will also be copied when the plasmid is copied.

There are a number of other features of plasmids that are useful in gene cloning. Naturally occurring plasmids can be quite large: some are more than 100 kb in size, but the ones used in routine gene cloning tend to be less than 10 kb. This makes them easy to purify and to manipulate. Plasmids commonly used in cloning contain a selectable marker, usually an antibiotic resistance gene. This means that you can tell which bacteria contain the plasmid simply by spreading them onto an agar plate containing the antibiotic. Those that contain the plasmid will grow and eventually

Box 3.1 Plasmid Origins of Replication

Naturally occurring plasmids have been modified by molecular biologists to produce the vectors that we use in gene cloning. Plasmids fall into a number of groups called incompatibility groups, depending on where the origin of replication of the plasmids is derived from. Plasmid replication uses many proteins, some encoded by the host bacterium, and some encoded by the plasmid; genes for the latter are often clustered near the origin of replication in the so-called *ori* region. Sequences in the *ori* region also control how many copies of the plasmid there are for each copy of the host chromosome.

Many of the plasmids used in gene cloning, such as pBR322, are based on the ColE1 origin of replication, so-called because they were originally derived from a plasmid called ColE1. Plasmids based on the ColE1 origin of replication are usually medium copy number plasmids with between 12 and 20 copies of the plasmid per cell. The plasmid pUC18 has the ColE1 origin but also has a mutation in the *ori* region which disrupts the mechanism that controls the copy number. The consequence of this is that there can be hundreds of copies of pUC18 for each copy of the chromosome. The ColE1 origin of replication only functions in *E. coli* and closely related species; these plasmids are said to have a narrow host range.

Plasmids based on origins which function in a range of organisms are described as broad host range plasmids. The two main examples here are the R (resistance) and the F (fertility) plasmids. In addition to having a broad host range, these plasmids have a low copy number with between two and five copies for the R plasmids and as few as one copy of the F plasmid per copy of the chromosome.